

CINEMA BOOKING SYSTEM



PREPARED BY:

Alraha abdullah

Halimah Ahmed alsfsafi

laila Omar alshulii

bayan obaid alrashdi

Shatha Abdullah Almarhabi



PROJECT DESCRIPTION

The Cinema Booking System is a database designed to manage cinema information, including movies, customers, halls, shows, and bookings. It provides a structured way to store and organize data through entities, attributes, and relationships.

SCENARIO

A cinema system is used to manage movies, customers, and bookings.

Below are the system details:

- A movie has a MovieID, title, genre, duration, and rating. Each movie can be shown multiple times in different schedules.
- A customer has a unique CustomerID, name, phone number, email, and age. Customers can make multiple bookings for different shows.
- A hall has a HallID, name, capacity, type, and location. Each hall can host multiple movie shows.
- A show includes a ShowID, date, and time. Each show is scheduled for a specific movie and takes place in a specific hall
- A booking has a BookingID, seat number, and payment status. Customers can book one or more seats for a show, and each booking is recorded in the system.

INTRODUCTION

The Cinema Booking System is designed to manage movie screenings, halls, customers, and booking operations in a cinema.

The system allows customers to book seats for different movie shows. Each show is scheduled in a specific hall and at a specific time. The database stores information about movies, halls, customers, shows, and bookings.

The purpose of this system is to organize cinema operations, reduce booking errors, and improve customer experience.

ENTITIES

01

Movie
* MovieID
* Title
* Genre
* Duration
* Rating

02

Hall
* HallID
* HallName
* Capacity
* Type
* Location

03

Customer
* CustomerID
* Name
* PhoneNumber
* Email
* Age

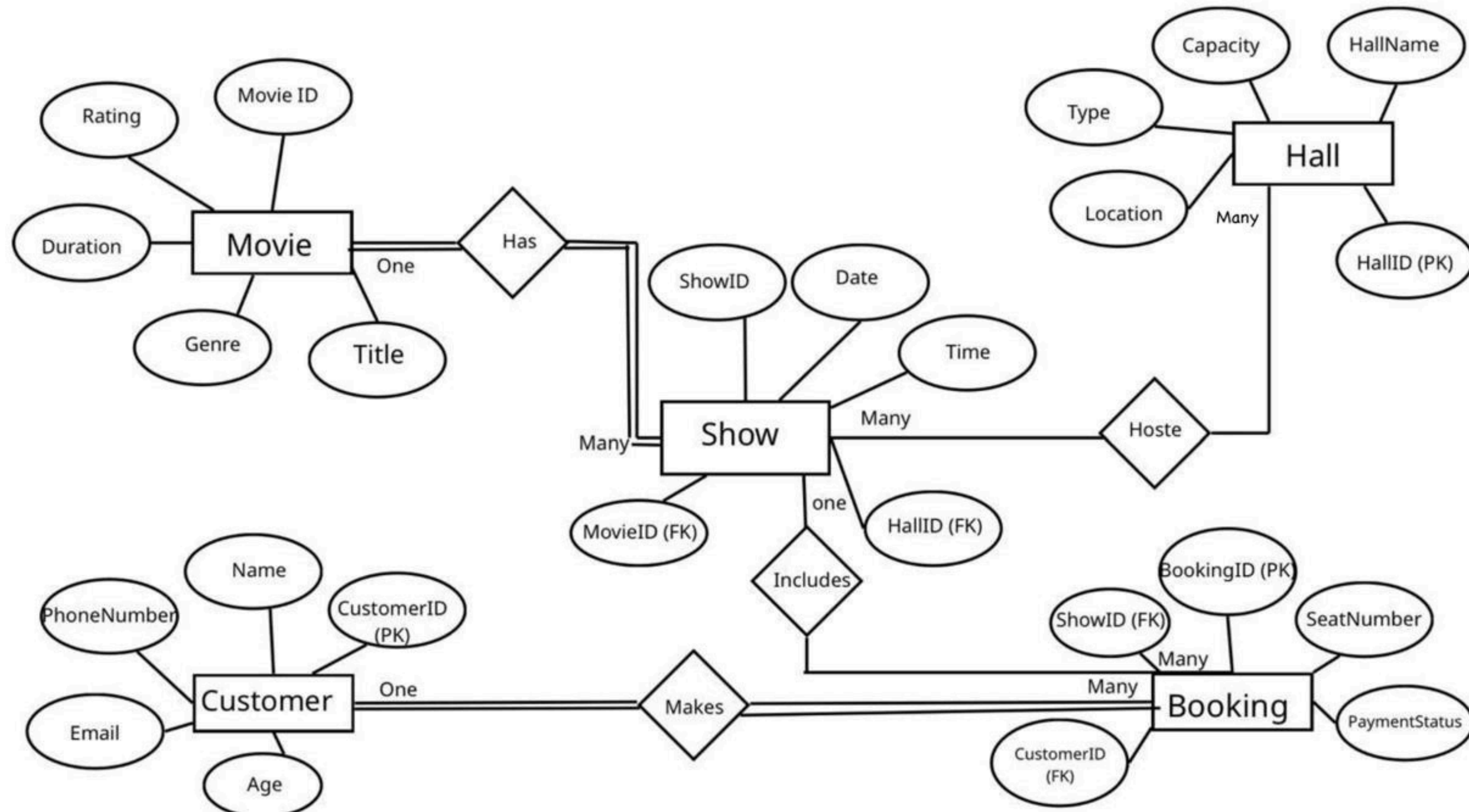
04

Booking
* BookingID
* PaymentStatus
* SeatNumber
* ShowID
* CustomerID

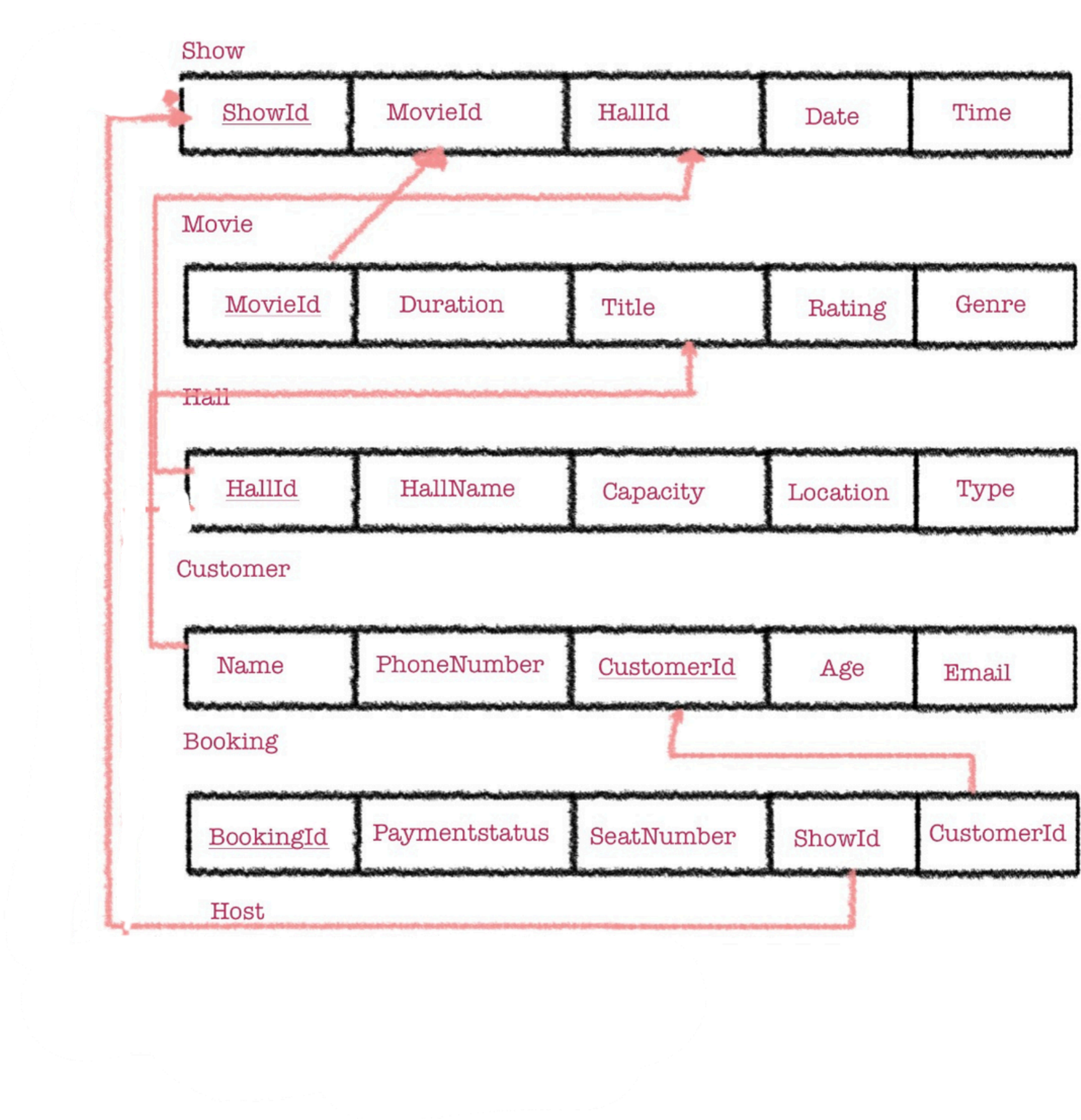
RELATIONSHIPS

- * One Movie can have many Shows.
- * One Hall can host many Shows.
- * One Show can have many Bookings.
- * One Customer can make many Bookings.
- * Each Booking belongs to one Customer and one Show.

ER DIAGRAM



SCHEMA



```
1 • DROP DATABASE IF EXISTS HomeTheater;
2 • CREATE DATABASE HomeTheater;
3 • USE HomeTheater;
4
5 • CREATE TABLE Movie (
6     MovieID INT PRIMARY KEY,
7     Title VARCHAR(100),
8     Genre VARCHAR(100),
9     Duration INT
10 );
11
12 • ALTER TABLE Movie
13 ADD Rating VARCHAR(20);
14
15 • INSERT INTO Movie VALUES
16     (1, 'Avengers Endgame', 'Action', 180, 'PG-13'),
17     (2, 'Frozen 2', 'Animation', 120, 'PG'),
18     (3, 'Joker', 'Drama', 122, 'R'),
19     (4, 'Inception', 'Sci-Fi', 148, 'PG-13'),
20     (5, 'Titanic', 'Romance', 195, 'PG-13'),
21     (6, 'Interstellar', 'Sci-Fi', 169, 'PG-13'),
22     (7, 'The Batman', 'Action', 176, 'PG-13'),
23     (8, 'Coco', 'Animation', 105, 'PG'),
24     (9, 'Avatar', 'Fantasy', 162, 'PG-13'),
25     (10, 'Moana', 'Adventure', 107, 'PG');
```

```
24      (9, 'Avatar', 'Fantasy', 162, 'PG-13'),
25      (10, 'Moana', 'Adventure', 107, 'PG');
26
27 • CREATE TABLE Hall (
28     HallID INT PRIMARY KEY,
29     HallName VARCHAR(100),
30     Capacity INT,
31     Type VARCHAR(50)
32 );
33
34 • ALTER TABLE Hall
35 ADD Location VARCHAR(100);
36
37 • INSERT INTO Hall VALUES
38     (1, 'Hall A', 100, 'VIP', 'First Floor'),
39     (2, 'Hall B', 150, 'Standard', 'Second Floor'),
40     (3, 'Hall C', 120, 'VIP', 'First Floor'),
41     (4, 'Hall D', 200, 'IMAX', 'Ground Floor'),
42     (5, 'Hall E', 90, 'Standard', 'Second Floor'),
43     (6, 'Hall F', 110, 'VIP', 'Third Floor'),
44     (7, 'Hall G', 130, 'IMAX', 'Ground Floor'),
45     (8, 'Hall H', 140, 'Standard', 'Second Floor'),
46     (9, 'Hall I', 160, 'VIP', 'Third Floor'),
47     (10, 'Hall J', 180, 'IMAX', 'Ground Floor');
48
```

```
48
49 • CREATE TABLE Shows (
50     ShowID INT PRIMARY KEY,
51     Date DATE,
52     MovieID INT,
53     HallID INT,
54     FOREIGN KEY (MovieID) REFERENCES Movie(MovieID),
55     FOREIGN KEY (HallID) REFERENCES Hall(HallID)
56 );
57
58 • ALTER TABLE Shows
59     ADD Time TIME;
60
61 • INSERT INTO Shows VALUES
62     (1, '2025-05-01', 1, 1, '18:00:00'),
63     (2, '2025-05-02', 2, 2, '20:00:00'),
64     (3, '2025-05-03', 3, 3, '16:30:00'),
65     (4, '2025-05-04', 4, 4, '19:00:00'),
66     (5, '2025-05-05', 5, 5, '21:00:00'),
67     (6, '2025-05-06', 6, 6, '17:00:00'),
68     (7, '2025-05-07', 7, 7, '22:00:00'),
69     (8, '2025-05-08', 8, 8, '15:00:00'),
70     (9, '2025-05-09', 9, 9, '18:30:00'),
71     (10, '2025-05-10', 10, 10, '20:30:00');
72
```

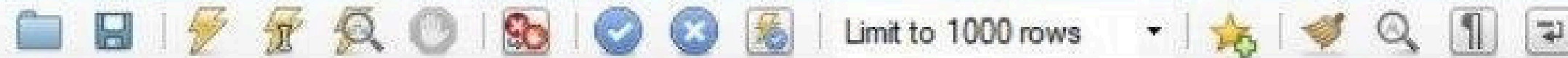
```
72
73 • CREATE TABLE Customer (
74     CustomerID INT PRIMARY KEY,
75     Name VARCHAR(100),
76     PhoneNumber VARCHAR(15),
77     Email VARCHAR(100)
78 );
79
80 • ALTER TABLE Customer
81 ADD Age INT;
82
83 • INSERT INTO Customer VALUES
84 (1, 'Ahmed Ali', '0551234567', 'ahmed@gmail.com', 22),
85 (2, 'Sara Khalid', '0569876543', 'sara@gmail.com', 25),
86 (3, 'Omar Hassan', '0543210987', 'omar@gmail.com', 30),
87 (4, 'Laila Ahmed', '0555678901', 'laila@gmail.com', 27),
88 (5, 'Fahad Ali', '0581239876', 'fahad@gmail.com', 35),
89 (6, 'Noura Saleh', '0568745123', 'noura@gmail.com', 28),
90 (7, 'Hassan Mohammed', '0531236547', 'hassan@gmail.com', 40),
91 (8, 'Reem Abdullah', '0598765432', 'reem@gmail.com', 24),
92 (9, 'Yousef Ibrahim', '0547893216', 'yousef@gmail.com', 31),
93 (10, 'Mona Saud', '0565432198', 'mona@gmail.com', 29);
94
95 • CREATE TABLE Booking (
96     BookingID INT PRIMARY KEY,
```

```
94
95 • CREATE TABLE Booking (
96     BookingID INT PRIMARY KEY,
97     PaymentStatus VARCHAR(50),
98     SeatNumber INT,
99     ShowID INT,
100     CustomerID INT,
101     FOREIGN KEY (ShowID) REFERENCES Shows(ShowID),
102     FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
103 );
104
105 • INSERT INTO Booking VALUES
106     (1, 'Paid', 15, 1, 1),
107     (2, 'Pending', 22, 2, 2),
108     (3, 'Paid', 10, 3, 3),
109     (4, 'Cancelled', 5, 4, 4),
110     (5, 'Paid', 18, 5, 5),
111     (6, 'Pending', 25, 6, 6),
112     (7, 'Paid', 12, 7, 7),
113     (8, 'Paid', 8, 8, 8),
114     (9, 'Cancelled', 30, 9, 9),
115     (10, 'Paid', 14, 10, 10);
116
117 • UPDATE Customer
118     SET Age = 26
```

```
116
117 • UPDATE Customer
118     SET Age = 26
119     WHERE CustomerID = 2;
120
121 • UPDATE Booking
122     SET PaymentStatus = 'Paid'
123     WHERE BookingID = 2;
124
125 • DELETE FROM Booking
126     WHERE BookingID = 9;
127
128 • SELECT Customer.Name, Movie.Title, Shows.Date, Booking.SeatNumber
129     FROM Booking
130     JOIN Customer
131     ON Booking.CustomerID = Customer.CustomerID
132     JOIN Shows
133     ON Booking.ShowID = Shows.ShowID
134     JOIN Movie
135     ON Shows.MovieID = Movie.MovieID;
136
137 • SELECT PaymentStatus, COUNT(BookingID) AS TotalBookings
138     FROM Booking
139     GROUP BY PaymentStatus;
140
```



```
137 • SELECT PaymentStatus, COUNT(BookingID) AS TotalBookings
138 FROM Booking
139 GROUP BY PaymentStatus;
140
141 • SELECT * FROM movie
142 ORDER BY Duration DESC;
143
144 • SELECT Hall.Type, COUNT(Shows.ShowID) AS NumberOfShows
145 FROM Shows
146 JOIN Hall
147 ON Shows.HallID = Hall.HallID
148 GROUP BY Hall.Type;
149
150 • SELECT Movie.Title, Hall.HallName, Shows.Time
151 FROM Shows
152 JOIN Movie
153 ON Shows.MovieID = Movie.MovieID
154 JOIN Hall
155 ON Shows.HallID = Hall.HallID;
156
157 • SELECT Customer.Name, Booking.PaymentStatus
158 FROM Booking
159 JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160 WHERE PaymentStatus = 'Paid';
```



```
153 ON Shows.MovieID = Movie.MovieID
154 JOIN Hall
155 ON Shows.HallID = Hall.HallID;
156
157 • SELECT Customer.Name, Booking.PaymentStatus
158 FROM Booking
159 JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160 WHERE PaymentStatus = 'Paid';
161
162 • SELECT * FROM Movie;
```

Result Grid



Filter Rows:

Edit:



Export/Import:

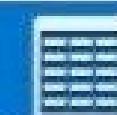
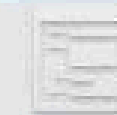
Wrap Cell Content: ☐Result
GridForm
EditorField
Types

	MovieID	Title	Genre	Duration	Rating
▶	1	Avengers Endgame	Action	180	PG-13
	2	Frozen 2	Animation	120	PG
	3	Joker	Drama	122	R
	4	Inception	Sci-Fi	148	PG-13
	5	Titanic	Romance	195	PG-13
	6	Interstellar	Sci-Fi	169	PG-13
	7	The Batman	Action	176	PG-13
	8	Coco	Animation	105	PG
	9	Avatar	Fantasy	162	PG-13
	10	Moana	Adventure	107	PG
•	NULL	NULL	NULL	NULL	NULL

```
153 ON Shows.MovieID = Movie.MovieID
154 JOIN Hall
155 ON Shows.HallID = Hall.HallID;
156
157 • SELECT Customer.Name, Booking.PaymentStatus
158 FROM Booking
159 JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160 WHERE PaymentStatus = 'Paid';
161
162 • SELECT * FROM Hall;
```

Result Grid   Filter Rows: Edit: Export/Import:  Wrap Cell Content: 

	HallID	HallName	Capacity	Type	Location
▶	1	Hall A	100	VIP	First Floor
	2	Hall B	150	Standard	Second Floor
	3	Hall C	120	VIP	First Floor
	4	Hall D	200	IMAX	Ground Floor
	5	Hall E	90	Standard	Second Floor
	6	Hall F	110	VIP	Third Floor
	7	Hall G	130	IMAX	Ground Floor
	8	Hall H	140	Standard	Second Floor
	9	Hall I	160	VIP	Third Floor
	10	Hall J	180	IMAX	Ground Floor
•	NULL	NULL	NULL	NULL	NULL


Result
Grid
Form
Editor
Field
Types



```
153 ON Shows.MovieID = Movie.MovieID
154 JOIN Hall
155 ON Shows.HallID = Hall.HallID;
156
157 • SELECT Customer.Name, Booking.PaymentStatus
158 FROM Booking
159 JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160 WHERE PaymentStatus = 'Paid';
161
162 • SELECT * FROM Shows;
```

Result Grid



Filter Rows:

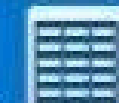
Edit:



Export/Import:



Wrap Cell Content:

Result
GridForm
EditorField
Types

	ShowID	Date	MovieID	HallID	Time
▶	1	2025-05-01	1	1	18:00:00
	2	2025-05-02	2	2	20:00:00
	3	2025-05-03	3	3	16:30:00
	4	2025-05-04	4	4	19:00:00
	5	2025-05-05	5	5	21:00:00
	6	2025-05-06	6	6	17:00:00
	7	2025-05-07	7	7	22:00:00
	8	2025-05-08	8	8	15:00:00
	9	2025-05-09	9	9	18:30:00
	10	2025-05-10	10	10	20:30:00
•	NULL	NULL	NULL	NULL	NULL

```
153 ON Shows.MovieID = Movie.MovieID
154 JOIN Hall
155 ON Shows.HallID = Hall.HallID;
156
157 • SELECT Customer.Name, Booking.PaymentStatus
158 FROM Booking
159 JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160 WHERE PaymentStatus = 'Paid';
161
162 • SELECT * FROM Customer;
```

Result Grid



Filter Rows:

Edit:



Export/Import:



Wrap Cell Content:

Result
GridForm
EditorField
Types

	CustomerID	Name	PhoneNumber	Email	Age
▶	1	Ahmed Ali	0551234567	ahmed@gmail.com	22
	2	Sara Khalid	0569876543	sara@gmail.com	26
	3	Omar Hassan	0543210987	omar@gmail.com	30
	4	Laila Ahmed	0555678901	laila@gmail.com	27
	5	Fahad Ali	0581239876	fahad@gmail.com	35
	6	Noura Saleh	0568745123	noura@gmail.com	28
	7	Hassan Mohammed	0531236547	hassan@gmail.com	40
	8	Reem Abdullah	0598765432	reem@gmail.com	24
	9	Yousef Ibrahim	0547893216	yousef@gmail.com	31
	10	Mona Saud	0565432198	mona@gmail.com	29
•	NULL	NULL	NULL	NULL	NULL



```
153  ON Shows.MovieID = Movie.MovieID
154  JOIN Hall
155  ON Shows.HallID = Hall.HallID;
156
157 •  SELECT Customer.Name, Booking.PaymentStatus
158  FROM Booking
159  JOIN Customer ON Booking.CustomerID = Customer.CustomerID
160  WHERE PaymentStatus = 'Paid';
161
162 •  SELECT * FROM Booking;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: ☐

	BookingID	PaymentStatus	SeatNumber	ShowID	CustomerID
▶	1	Paid	15	1	1
	2	Paid	22	2	2
	3	Paid	10	3	3
	4	Cancelled	5	4	4
	5	Paid	18	5	5
	6	Pending	25	6	6
	7	Paid	12	7	7
	8	Paid	8	8	8
	10	Paid	14	10	10
•	NULL	NULL	NULL	NULL	NULL



Result
Grid



Form
Editor



Field
Types

DIFFICULTIES we faced

- Managing multiple entities (movies, customers, halls, bookings) was challenging.
- Understanding relationships between entities was sometimes confusing.
- Designing the database schema correctly required careful planning.
- Limited time made it difficult to complete all tasks smoothly.

Task Distribution

alraha, halimah	Scenario Entities & Attributes Relationships & ER Diagram
byan	schema Diagram
shatha	Creating tables
Laila	SQL Manipulation Commands & SQL Complex Querying